Task 1 — Verify Kali user and network interfaces

Open Terminal and run:
- whoami

Then:
- ifconfig

Task 2 — Scan your own local network

First identify your Kali IP/subnet:
- ip addr

For example, if your Kali VM has:
- 192.168.29.65/24

scan only that authorized subnet:
- nmap -sn 192.168.29.0/24 -oN network_scan.txt

View the saved result:
- cat network_scan.txt

This performs host discovery and saves the results in:
- network_scan.txt

Task 3 — Hydra against your own SSH test VM

On the target VM:
- sudo systemctl status ssh

If needed:
- sudo systemctl start ssh

Check its IP:
- ip addr

Safe lab setup

Create a tiny wordlist containing only credentials you intentionally created for this lab:
- nano passwords.txt

Example:

- wrongpass
- test123
- LabPassword123

Then, from Kali, run Hydra against your own VM only:

- hydra -l testuser -P passwords.txt ssh://TARGET_IP

Replace:

- testuser

with your lab username and:

- TARGET_IP

with your own SSH VM's IP.

Example:

- hydra -l student -P passwords.txt ssh://192.168.29.70

Task 4 — Crack the tops123 test hash with John

Generate an MD5-crypt hash:

- openssl passwd -1 tops123

You will receive a hash similar to:

- 1randomsalt$hashedvalue

Copy that output into a file:

- nano hash.txt

Then create a small wordlist for the lab:

- nano wordlist.txt

Add:

- password
- 123456
- tops123
- admin123

Run John:

- john --wordlist=wordlist.txt hash.txt

View the recovered password:

- john --show hash.txt

Expected result:
- tops123

Task 5 — Basic Linux privilege-escalation enumeration

Step 1: Identify the current user
- whoami
- id

Step 2: Check available sudo permissions
- sudo -l

Look for commands the current user may run with elevated privileges.
Step 3: Check system information
- uname -a
- cat /etc/os-release

Step 4: Look for SUID files
- find / -perm -4000 -type f 2>/dev/null

Step 5: Look for writable files and directories
- find / -writable -type d 2>/dev/null

Step 6: Check scheduled tasks
- cat /etc/crontab
- ls -la /etc/cron.*

Step 7: Check running processes
- ps aux

Step 8: Check network services
- ss -tulpn

Comparison table

AI Suggested Tool/Command	Available?	Result
nmap	Check with which nmap	Network enumeration
hydra	Check with which hydra	Authentication testing
john	Check with which john	Password auditing

linpeas	May not be preinstalled	Linux privilege-escalation enumeration
sudo -l	Built-in system command	Checks sudo permissions
find	Built-in/common command	Finds SUID/writable files